UNIVERSITE DE FIANARANTSOA

ECOLE NATIONALE D'INFORMATIQUE

Rapport de Projet Routage IP

Mention : Informatique

Parcours : Informatique Générale

INTITULE :

Etude et Redistribution de Protocole de Routage RIP version 2 et OSPF

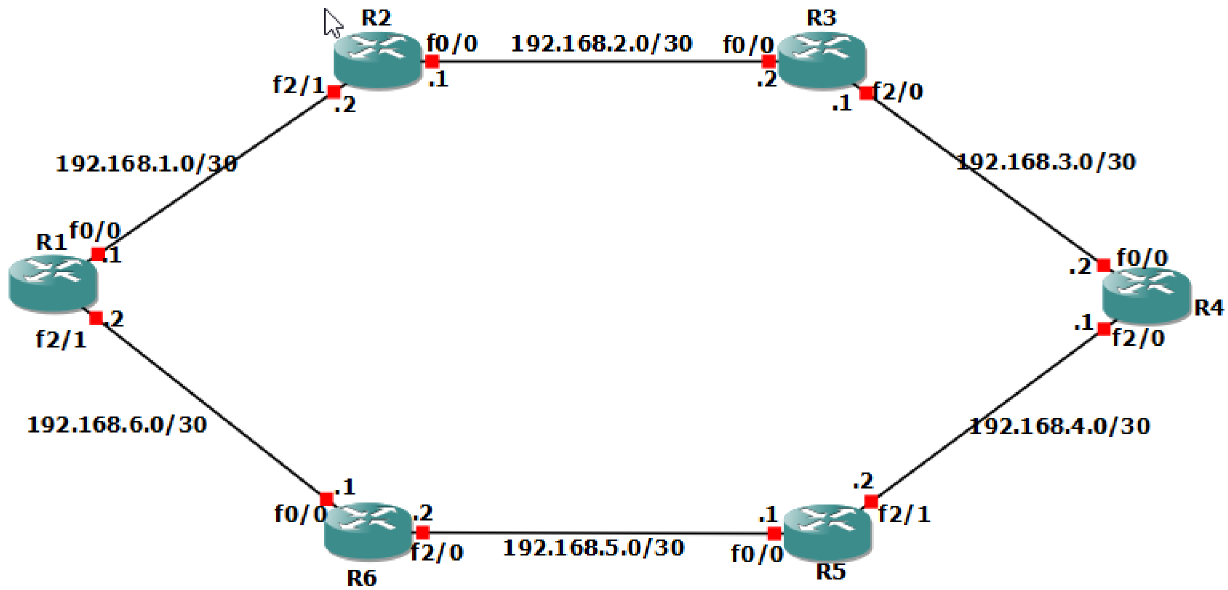
Présenté par :

- RATEFISON Yakin Ny Aina — Matricule : 2205 H-F
- ANDRIANIRINA Henikaja David — Matricule : 2267 H-F
- MIANDRISOA FANILOMANDRESY Orlando — Matricule : 2272 H-F
- RAVALISON RABOANA Andoniaina Fenosoa — Matricule : 2303 H-F

Enseignent :

- Mr RAZAFINDRAMONJA Clément Aubert

TOPOLOGIE DE LA PARTIE RIP :



❖ CONFIGURATION DES ROUTEURS

Configuration de chaque routeur et activation de RIPv2 , et on va le commencer par le routeur R1 . Tous les routeurs sont tous configurés selon leurs cas, ici on parle d'abord du **RIP.**

CONFIGUTRATION DU ROUTEUR R1 :

La configuration du routeur est présentées ci-dessous :

R1 R2 R3 R4 R5 R6 R7 R8 R9

```
*Apr 30 09:00:32.427: %LINK-5-CHANGED: Interface Ethernet1/4, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/5, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/6, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/7, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip add 192.168.1.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
*Apr 30 09:01:41.663: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:01:42.663: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int f2/1
R1(config-if)#ip add 192.168.6.2 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Apr 30 09:02:21.175: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:02:22.175: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.6.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Apr 30 09:57:58.827: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
Building configuration...
[OK]
R1#
```

solarwinds | Solar-PuTTY free tool

CONFIGURATION DU ROUTEUR R2 :

R1 R2 R3 R4 R5 R6 R7 R8 R9

```
*Apr 30 09:04:18.127: %LINK-5-CHANGED: Interface Ethernet1/5, changed state to administratively down
*Apr 30 09:04:18.139: %LINK-5-CHANGED: Interface Ethernet1/6, changed state to administratively down
*Apr 30 09:04:18.143: %LINK-5-CHANGED: Interface Ethernet1/7, changed state to administratively down
*Apr 30 09:04:18.167: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
*Apr 30 09:04:18.171: %LINK-5-CHANGED: Interface FastEthernet2/1, changed state to administratively down
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip add 192.168.2.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int
*Apr 30 09:06:38.415: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:06:39.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#int f2/1
R2(config-if)#ip add 192.168.1.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Apr 30 09:07:16.795: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:07:17.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 192.168.1.0
R2(config-router)#network 192.168.2.0
R2(config-router)#end
R2#
*Apr 30 09:57:46.887: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R2#
```

solarwinds | Solar-PuTTY free tool

CONFIGURATION DU ROUTEUR R3 :


```
*Apr 30 09:09:20.327: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/4, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/5, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)#ip add 192.168.2.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Apr 30 09:10:11.075: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:10:12.075: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#int f2/0
R3(config-if)#ip add 192.168.3.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
*Apr 30 09:10:53.275: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:10:54.275: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R3(config-if)#exit
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.2.0
R3(config-router)#network 192.168.3.0
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
*Apr 30 09:57:30.567: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R3#
```

CONFIGURATION DU ROUTEUR R4 :

```
*Apr 30 09:12:20.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip add 192.168.3.2 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:08.539: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:16:09.539: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#int f2/0
R4(config-if)#ip add 192.168.4.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:58.587: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:16:59.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R4(config)#int f2/1
R4(config-if)#ip add 172.16.10.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:17:49.963: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:17:50.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
R4(config-router)#
```

CONFIGURATION DU ROUTEUR R5 :

```
R1  R2  R3  R4  R5  R6  R7  R8  R9

*Apr 30 09:22:06.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:22:06.763: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:22:06.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
*Apr 30 09:22:06.783: %LINEPROTO-5-UPDOWN: Line pro
R5#cotocol on Interface FastEthernet2/1, changed state to down
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#int f2/1
R5(config-if)#ip add 192.168.4.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Apr 30 09:22:54.487: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:22:55.487: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R5(config)#int f0/0
R5(config-if)#ip add 192.168.5.1 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#
*Apr 30 09:23:38.943: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:23:39.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5(config-if)#exit
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#network 192.168.4.0
R5(config-router)#network 192.168.5.0
R5(config-router)#end
R5#wr
*Apr 30 09:58:08.607: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R5#

solarwinds | Solar-PuTTY free tool          © 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.
```

CONFIGURATION DU ROUTEUR R6 :

```
R1  R2  R3  R4  R5  R6  R7  R8  R9
*Apr 30 09:24:49.939: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/4, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/5, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#int f2/0
R6(config-if)#ip add 192.168.5.2 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Apr 30 09:25:40.283: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:25:41.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R6(config)#int f0/0
R6(config-if)#ip add 192.168.6.1 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
*Apr 30 09:26:35.511: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:26:36.511: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R6(config-if)#exit
R6(config)#router rip
R6(config-router)#version 2
R6(config-router)#no auto-summary
R6(config-router)#network 192.168.5.0
R6(config-router)#network 192.168.6.0
R6(config-router)#end
R6#w
*Apr 30 09:58:19.567: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R6#
solarwinds | Solar-PuTTY free tool    © 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.
```

❖ OBSERVATION DES TABLES DE ROUTAGE :

Dans cette section, nous allons examiner les tables de routage afin de mieux comprendre le fonctionnement des réseaux et la circulation des données, permettant d'identifier les chemins empruntés par les paquets dans un réseau.

Table de routage de routeur R1 :

Voici la présentation de la table de routage du routeur R1 pour plus d'explication sur ce routeur. Elle est montrées ci-dessous :

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                     [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/30 is directly connected, FastEthernet2/1
L        192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
```

- Passons maintenant au routeur R2.

Table de routage du routeur R2 :

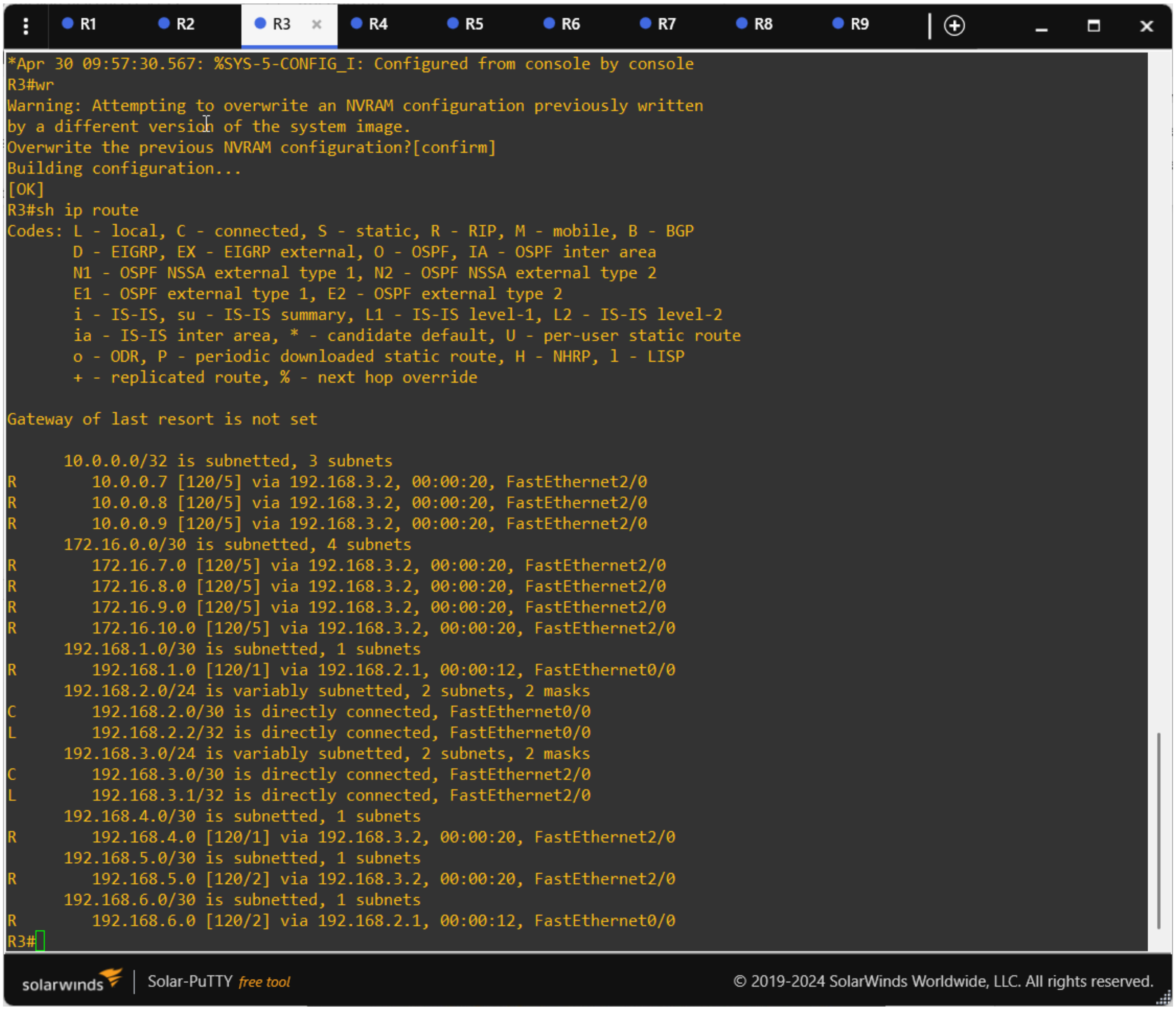
Présentation de la table de routage R2, elle montre les chemins utilisés pour atteindre les différents réseaux.

```
[OK]
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        10.0.0.8 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        10.0.0.9 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.8.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.9.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.10.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet2/1
L        192.168.1.2/32 is directly connected, FastEthernet2/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, FastEthernet0/0
L        192.168.2.1/32 is directly connected, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/1] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/2] via 192.168.1.1, 00:00:09, FastEthernet2/1
      192.168.6.0/30 is subnetted, 1 subnets
R        192.168.6.0 [120/1] via 192.168.1.1, 00:00:09, FastEthernet2/1
R2#
```

La table de routage de R2 est bien en place pour voir les adressages vers les autres routeurs.

Tables de routage du routeur R3 :

Dans cette table de routage, ellle met en évidence les réseaux directement connectés ainsi que les routes apprises via le protocole RIP, notammnet a travers les routeurs R2 et R4.



```
*Apr 30 09:57:30.567: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R        10.0.0.8 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R        10.0.0.9 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R        172.16.8.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R        172.16.9.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R        172.16.10.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
      192.168.1.0/30 is subnetted, 1 subnets
R        192.168.1.0 [120/1] via 192.168.2.1, 00:00:12, FastEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, FastEthernet0/0
L        192.168.2.2/32 is directly connected, FastEthernet0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, FastEthernet2/0
L        192.168.3.1/32 is directly connected, FastEthernet2/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/1] via 192.168.3.2, 00:00:20, FastEthernet2/0
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/2] via 192.168.3.2, 00:00:20, FastEthernet2/0
      192.168.6.0/30 is subnetted, 1 subnets
R        192.168.6.0 [120/2] via 192.168.2.1, 00:00:12, FastEthernet0/0
R3#
```


Tables de routages du routeur R4 :

Le routeur R4 regroupe les informations d'acheminement issues de deux protocoles de routage différents (RIP et OSPF). Il est le point d'interconnexion entre ces deux domaines, notamment par le routeur R7.

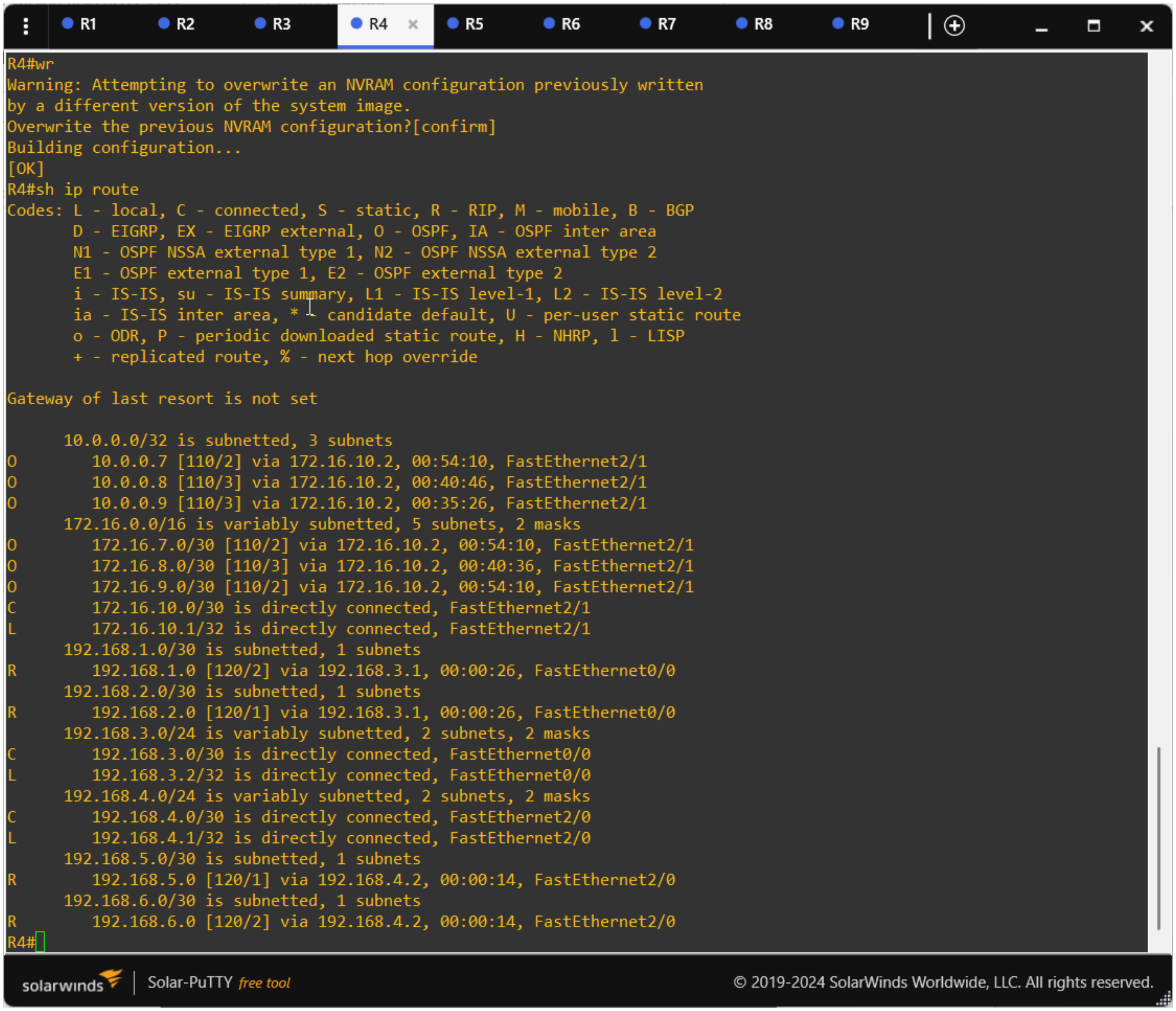
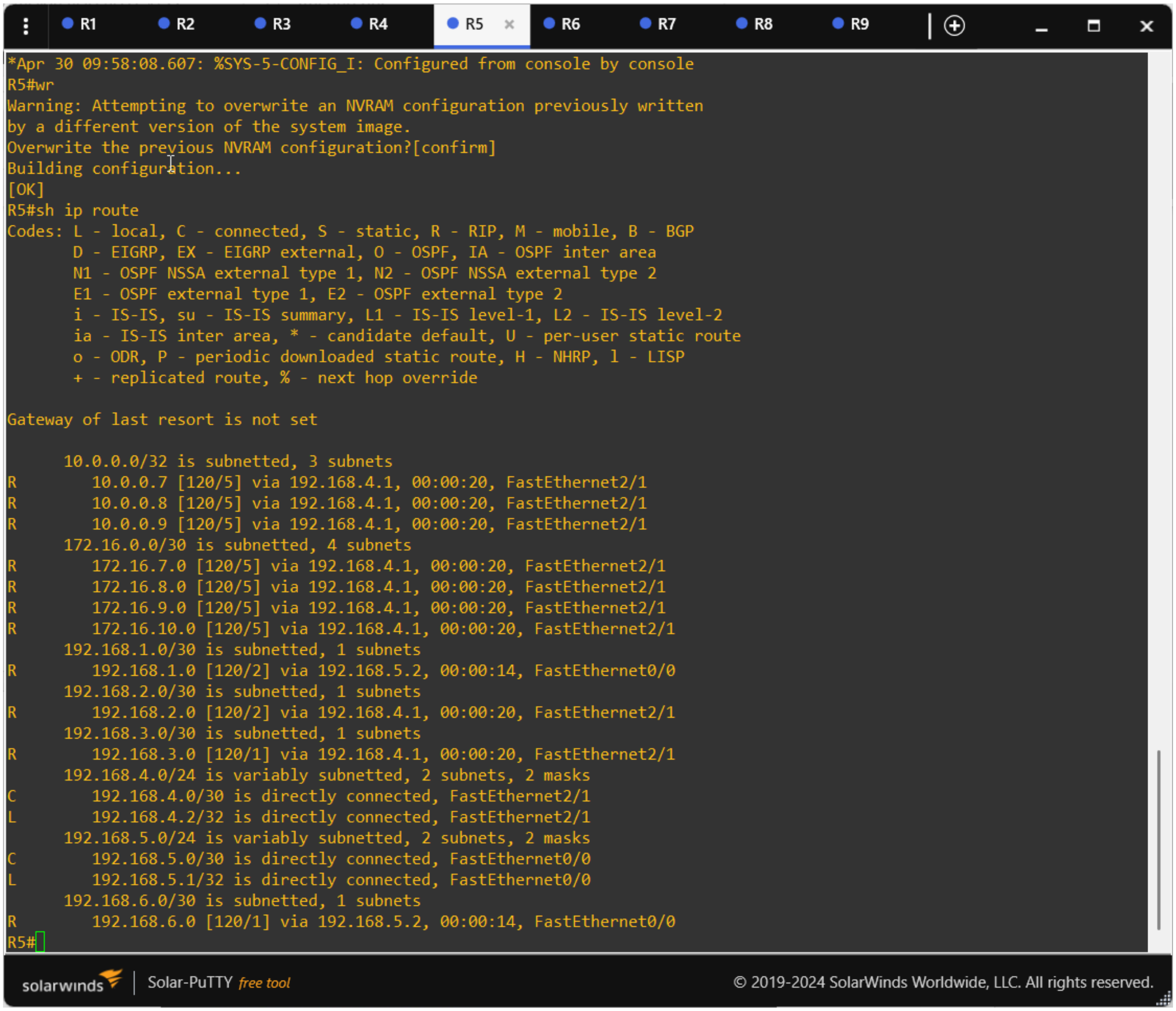


Table de routage du routeur R5 :

Il met en évidence les chemins de transmission passant par les routeurs R4 et R6 pour atteindre les les autres segments du réseaux.



```
*Apr 30 09:58:08.607: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R        10.0.0.8 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R        10.0.0.9 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R        172.16.8.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R        172.16.9.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R        172.16.10.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
      192.168.1.0/30 is subnetted, 1 subnets
R        192.168.1.0 [120/2] via 192.168.5.2, 00:00:14, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/2] via 192.168.4.1, 00:00:20, FastEthernet2/1
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/1] via 192.168.4.1, 00:00:20, FastEthernet2/1
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/30 is directly connected, FastEthernet2/1
L        192.168.4.2/32 is directly connected, FastEthernet2/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, FastEthernet0/0
L        192.168.5.1/32 is directly connected, FastEthernet0/0
      192.168.6.0/30 is subnetted, 1 subnets
R        192.168.6.0 [120/1] via 192.168.5.2, 00:00:14, FastEthernet0/0
R5#
```


Table de routage du routeur R6 :

Il a une fonction dans la continuité du routage au sein du domaine RIP et permet d'identifier les réseaux accessibles ainsi que les routes échangées avec les routeurs voisins (R5 et R1) .

R1 R2 R3 R4 R5 R6 R7 R8 R9

```
R6#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R        10.0.0.8 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R        10.0.0.9 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R        172.16.8.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R        172.16.9.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R        172.16.10.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
      192.168.1.0/30 is subnetted, 1 subnets
R        192.168.1.0 [120/1] via 192.168.6.2, 00:00:23, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/2] via 192.168.6.2, 00:00:23, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/2] via 192.168.5.1, 00:00:28, FastEthernet2/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/1] via 192.168.5.1, 00:00:28, FastEthernet2/0
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, FastEthernet2/0
L        192.168.5.2/32 is directly connected, FastEthernet2/0
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/30 is directly connected, FastEthernet0/0
L        192.168.6.1/32 is directly connected, FastEthernet0/0
R6#
R6#
```

solarwinds | Solar-PuTTY *free tool*

Dans la prochaine page, on va faire une analyse sur un routeur pour voir plus de details au routage.

❖ CONTENU D’UNE TABLE DE ROUTAGE :

• Routeur R1 :

La table de routage de R1 contient les informations suivantes :

- Le réseau 192.168.1.0/30 est directement connecté via l'interface FastEthernet0/0, avec une adresse de passerelle (gateway) de 192.168.1.1

- Le réseau 192.168.1.4/30 est atteint via une route apprise par le protocole de routage avec une métrique de 120 et la passerelle est 192.168.1.2 via l'interface FastEthernet0/0.

- Le réseau 192.168.1.8/30 est atteint via la même passerelle 192.168.1.2 avec une métrique de 120.

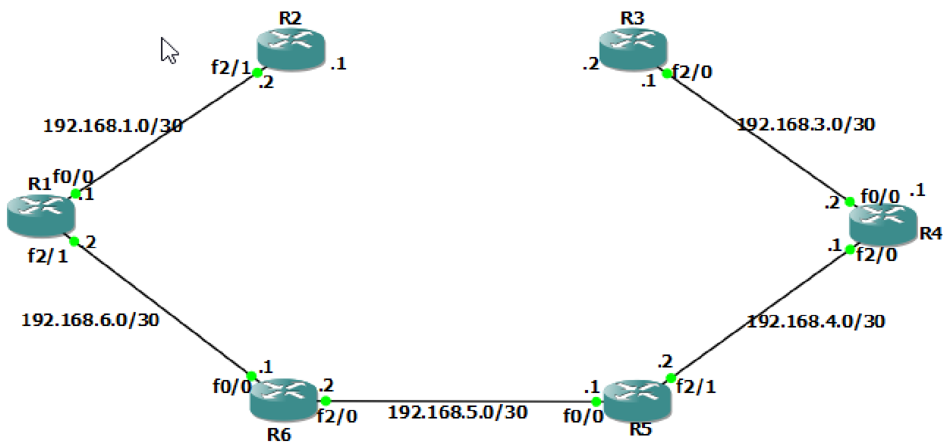
- Le réseau 192.168.1.12/30 est atteint via une autre passerelle 192.168.1.22 via l'interface Ethernet1/0 avec une métrique de 120.

- Le réseau 192.168.1.16/30 est atteint également via la passerelle 192.168.1.0 avec une métrique de 120.

- Le réseau 192.168.1.20/30 est directement connecté via l'interface Ethernet1/0.

- Enfin, il y a une route locale (L) pour chaque adresse IP des interfaces FastEthernet0/0 et Ethernet1/0 (192.168.1.1 et 192.168.1.21 respectivement).

❖ SUPPRESSION DE LA LIAISON ENTRE R2 ET R3 ET DESACTIVATION DES INTERFACES CORRESPONDANTES :



• Temps de jonction des routeurs :

Dans RIP, le routeur R1 peut joindre R3 dans un temps typique jusqu' à 180 secondes.

```
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#
```

❖ ENCAPSULATIONS DES PAQUETS **RIP**

Avec le protocole de routage RIP v2, le temps nécessaire à R1 pour détecter les changements dans la topologie du réseau et mettre à jour sa table de routage dépendra des paramètres de temporisation RIP v2 configurés sur les routeurs.

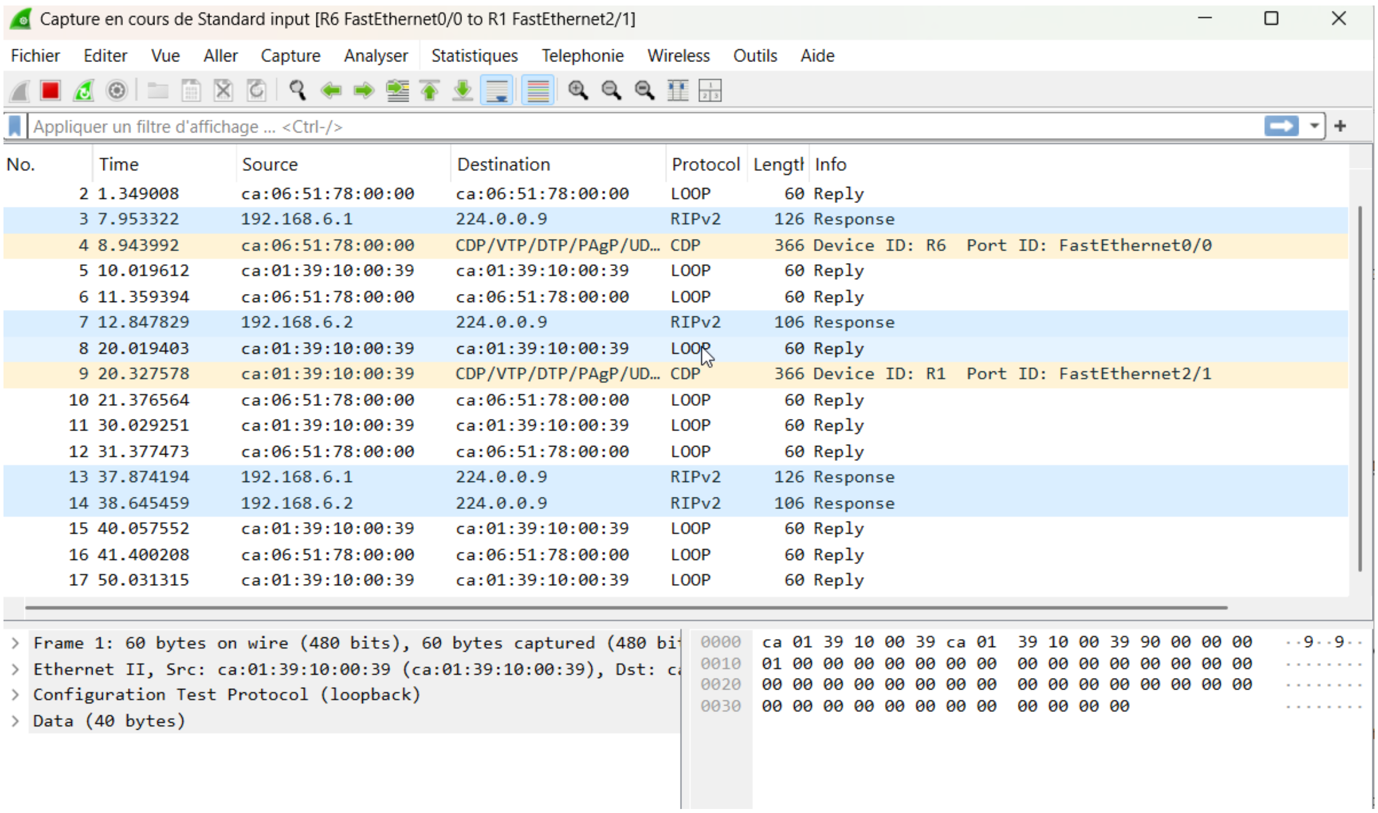
Par défaut, RIP v2 utilise une temporisation de mise à jour de 30 secondes, ce qui signifie que R1 enverra des mises à jour de sa table de routage à ses voisins RIP toutes les 30 secondes. Après la suppression du lien entre R2 et R3 et la désactivation des interfaces correspondantes sur le réseau 192.168.2.0/30, R1 devra attendre 120s pour atteindre le router R3.

Une fois que R1 détecte le changement, il commencera le processus de convergence RIP, qui peut prendre plusieurs mises à jour du protocole avant que la nouvelle topologie soit pleinement reflétée dans sa table de routage.

• Numéro de Port :

Les paquets RIP sont encapsulé dans le Protocole UDP, numero de port 520.

❖ REMISE DE LIEN ENTRE R2 ET R3 :



- ### Observation :

-Il existe des nouveaux paquets envoyés entre le routeur R2 et R3.
-Il y a un changement du comportement des réseaux.

- ### Commentaires :

Request/Response : Ces messages sont utilisés pour demander ou répondre aux mises à jour périodiques de la table de routage. Chaque routeur envoie périodiquement un message Request pour demander des mises à jour de ses voisins (avec l'adresse multicast 224.0.0.9 et le port UDP 520). Les voisins répondent ensuite avec des messages Response contenant les informations de leur table de routage.

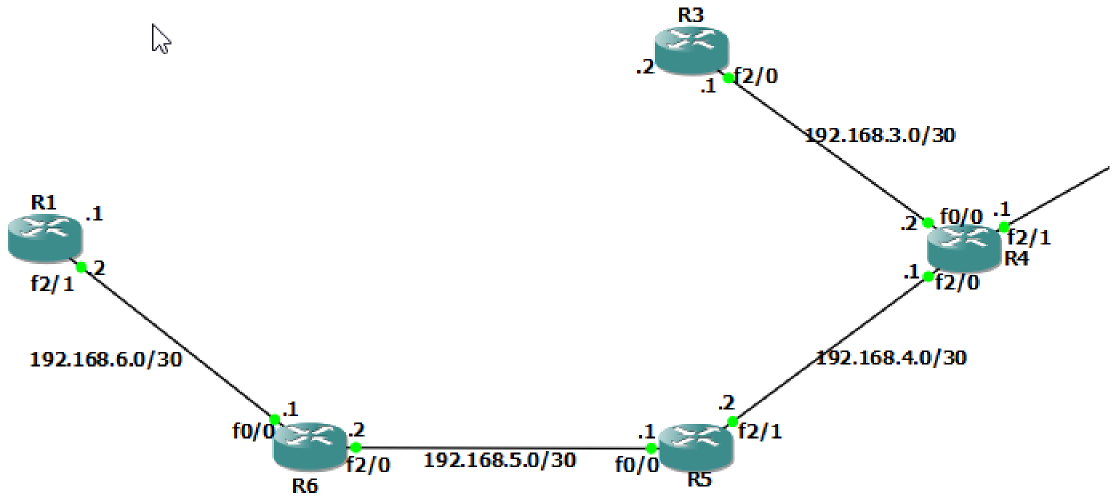
Il y a donc des échanges de message RIPv2 entre les routeurs en utilisant l'adrsesse multicast ci-dessus.

En cas de rupture de liaison, les messages disparaitraient et entrainant une convergence du réseau et suppression des routes invalides.

❖ SUPPRESSION DU ROUTEUR R2

Entrainant la perte des échanges de mises à jour de routage ;

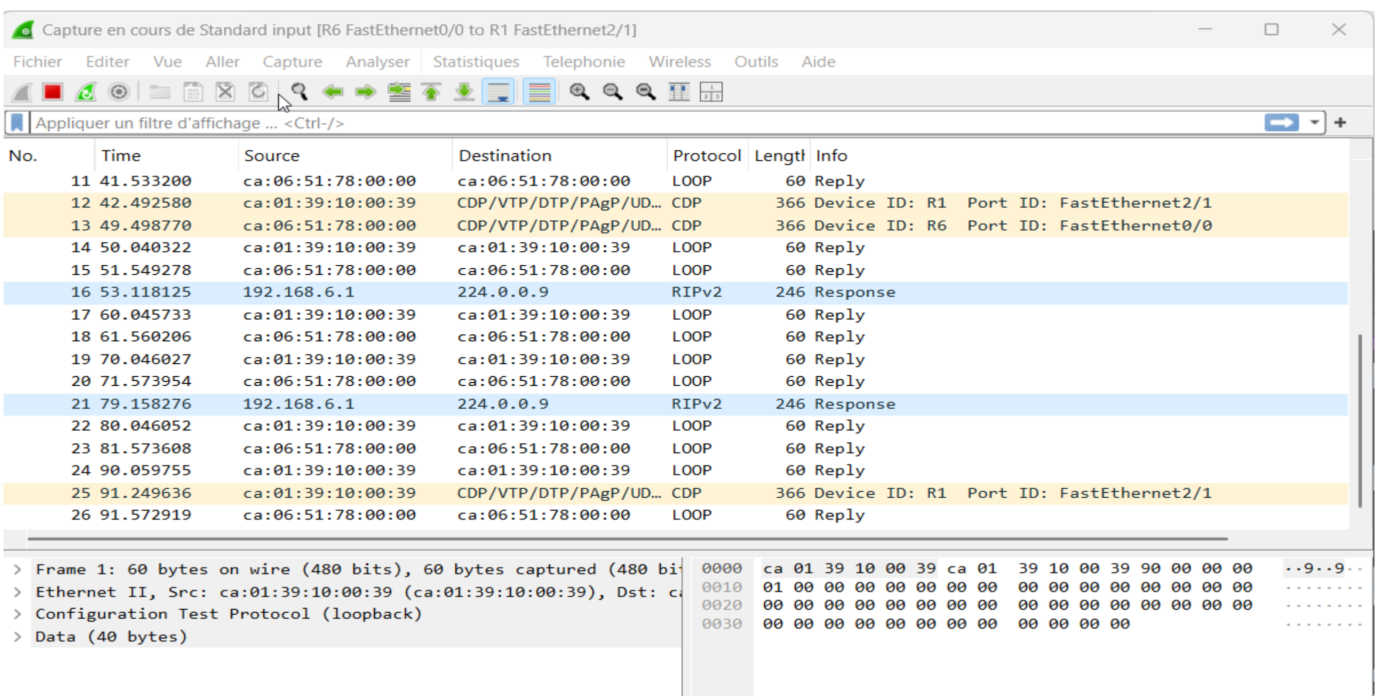
Présentation de la suppression :



Routeur R2 supprimé, ce qui entraîne la suppression des réseaux 192.168.1.0 et 192.168.1.2, cela déclenchera une convergence du protocole de routage RIP sur les routeurs R1 et R3.

• Messages échangés :

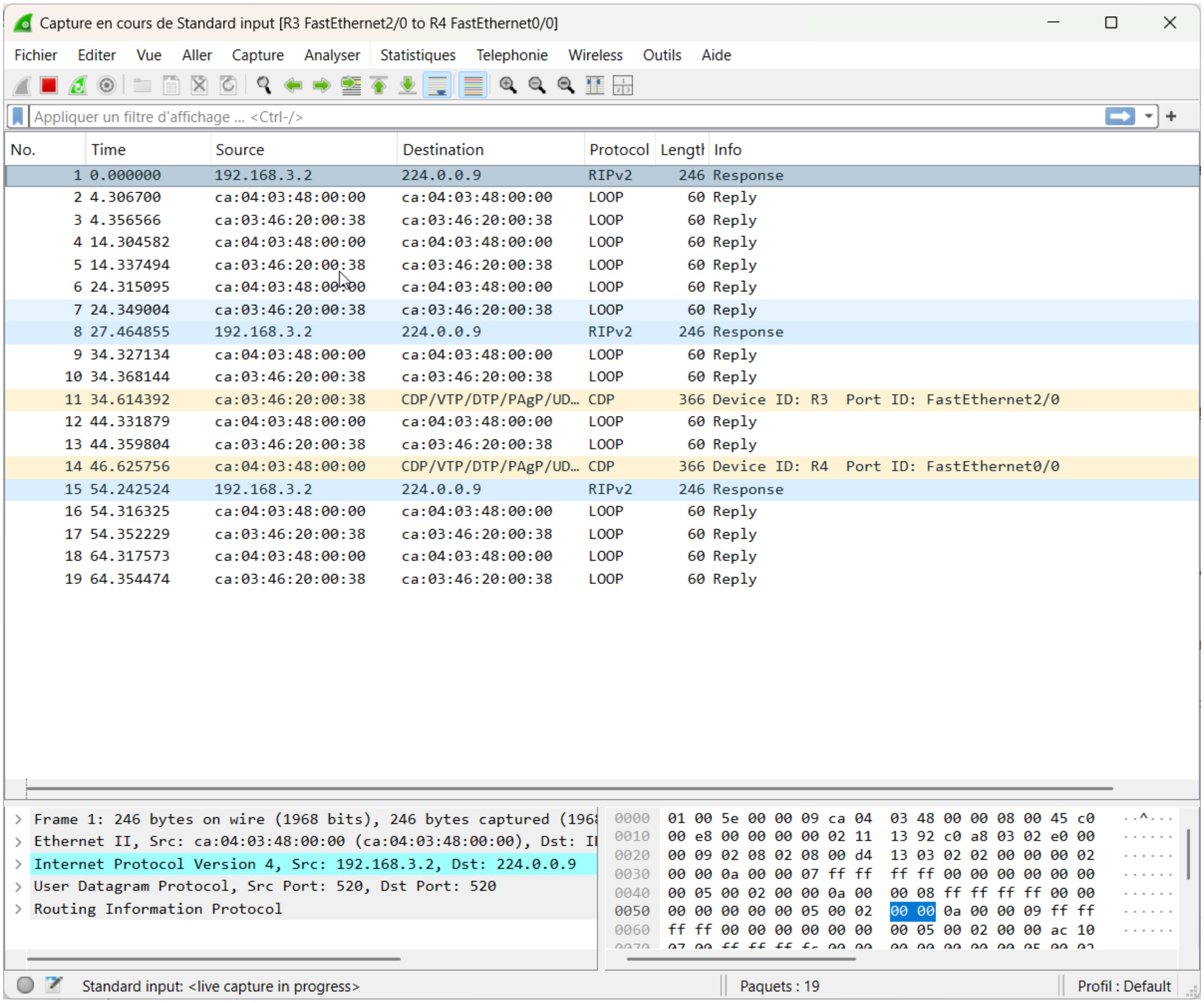
Dans Wireshark,



Sortie de R1 :

-R1 enverra des messages RIP Response pour informer ses voisins de la suppression des routes vers les réseaux 192.168.1.0 et 192.168.1.2.

-R1 recevra également des messages RIP Update des autres routeurs, probablement R6, pour indiquer la même chose.



Sortie de R3 :

-R3 enverra des messages RIP Response pour informer ses voisins de la suppression des routes vers les réseaux 192.168.1.0 et 192.168.1.2.

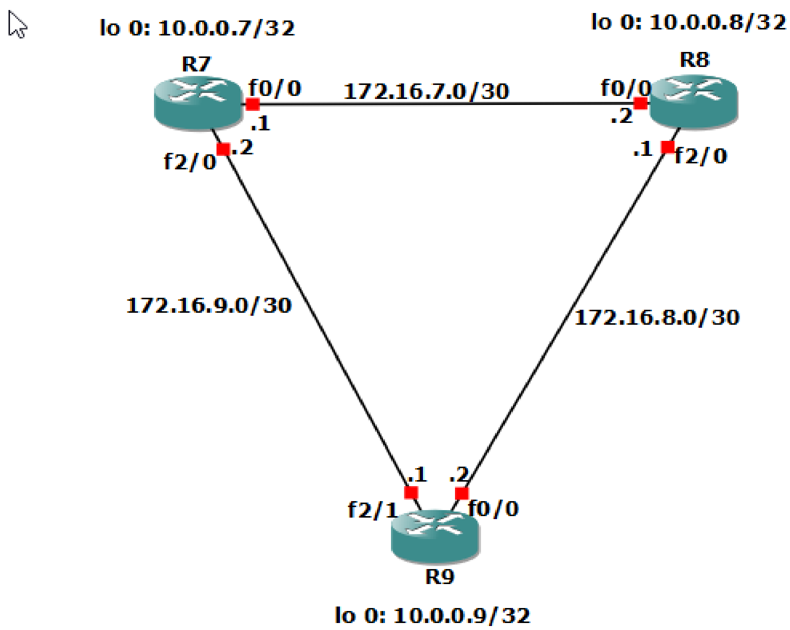
-R3 recevra également des messages RIP Update des autres routeurs, probablement R4, pour indiquer la même chose.

- ## Temps de convergence :

La convergence peut prendre vers 180 secondes après la considération que le routeur R2 est en panne.

Partie OSPF et Redistribution

➢ Voici la topologie OSPF :



❖ CONFIGURATION DES ROUTEURS ET ACTIVATION DES INTERFACES DU RESEAU

Configuration des routeurs de la partie OSPF, on va voir toutes les configurations des routeurs (R7, R8 et R9) ci-après avec l'activation des interfaces.

CONFIGURATION DU ROUTEUR R7 :

R1 R2 R3 R4 R5 R6 R7 R8 R9

```
R7#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#int lo 0
R7(config-if)#
*Apr 30 09:28:21.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R7(config-if)#ip add 10.0.0.7 255.255.255.255
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#int f2/1
R7(config-if)#ip add 172.16.10.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:30:34.367: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:30:35.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R7(config)#int f0/0
R7(config-if)#ip add 172.16.7.1 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
*Apr 30 09:31:32.599: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:31:33.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7(config-if)#exit
R7(config)#int f2/0
R7(config-if)#ip add 172.16.9.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:32:50.631: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
R7(config)#
*Apr 30 09:32:51.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
```

solarwinds | Solar-PuTTY *free tool*


```
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
R7(config-router)#
*Apr 30 09:35:13.879: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet2/1 from LOADING to FULL, Loading Done
R7(config-router)#exit
R7(config)#redistribute rip subnets metric 20 metric-type 2
               ^
% Invalid input detected at '^' marker.

R7(config)#redistribute rip subnets metric 20 metric-type 2
               ^
% Invalid input detected at '^' marker.

R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
R7(config-router)#redistribute rip subnets metric 20 metric-type 2
R7(config-router)#exit
R7(config)#router rip
```

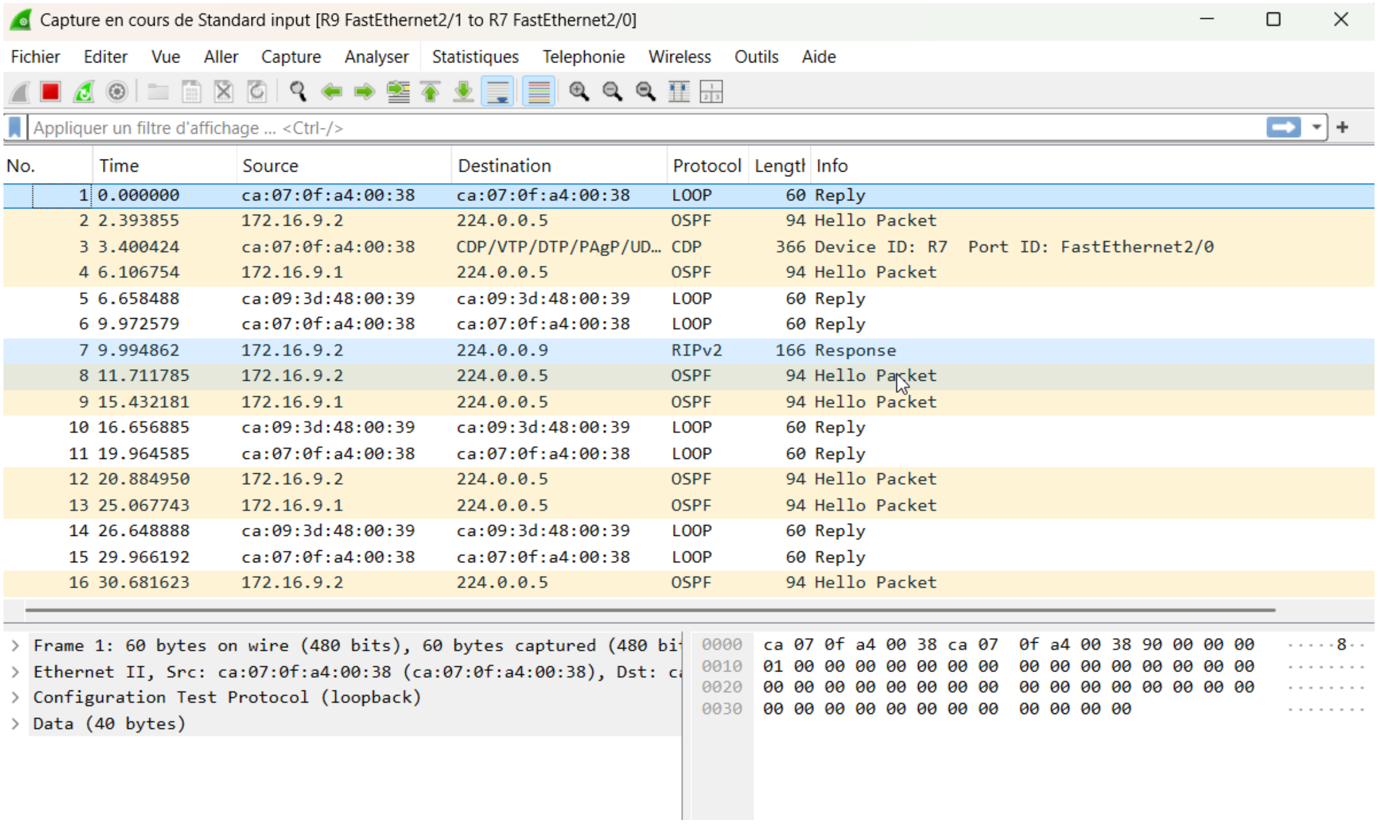
CONFIGURATION DU ROUTEUR R8 :

```
R8#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#int lo 0
R8(config-if)#
*Apr 30 09:44:01.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R8(config-if)#ip add 10.0.0.8 255.255.255.255
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#int f0/0
R8(config-if)#ip add 172.16.7.2 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Apr 30 09:45:54.059: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:45:55.059: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R8(config)#int f2/0
R8(config-if)#ip add 172.16.8.1 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
*Apr 30 09:46:38.195: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:46:39.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R8(config-if)#exit
R8(config)#router ospf 1
R8(config-router)#router-id 8.8.8.8
R8(config-router)#network 10.0.0.8 0.0.0.0 area 0
R8(config-router)#network 172.16.7.0 0.0.0.3 area 0
R8(config-router)#network 172.16.8.0 0.0.0.3 area 0
*Apr 30 09:48:37.539: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet0/0 from LOADING to FULL, Loading Done
R8(config-router)#network 172.16.8.0 0.0.0.3 area 0
R8(config-router)#
*Apr 30 09:54:36.267: %OSPF-5-ADJCHG: Process 1, Nbr 9.9.9.9 on FastEthernet2/0 from LOADING to FULL, Loading Done
R8(config-router)#end
R8#w
*Apr 30 09:58:40.007: %SYS-5-CONFIG_I: Configured from console by console
R8#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
```

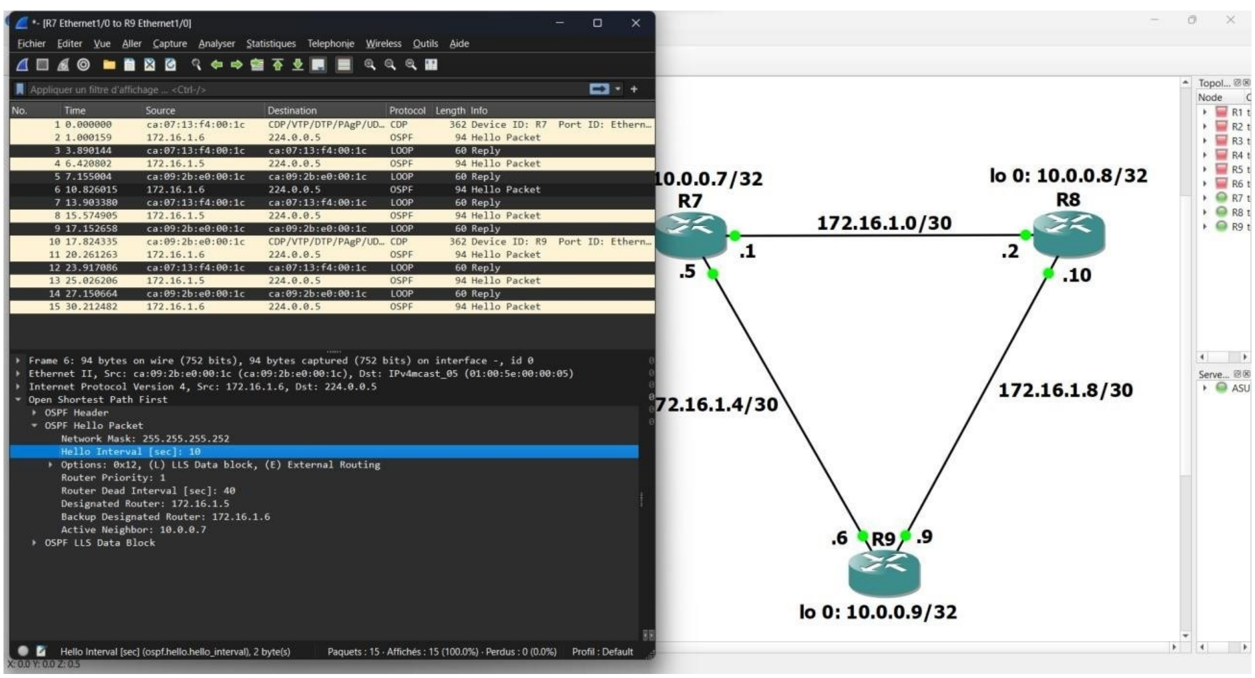

CONFIGURATION DU ROUTEUR R9 :

Toutes les configurations des routeurs dans l'OSPF sont tous faites.

❖ MESSAGES ECHANGES ENTRE LES ROUTEURS

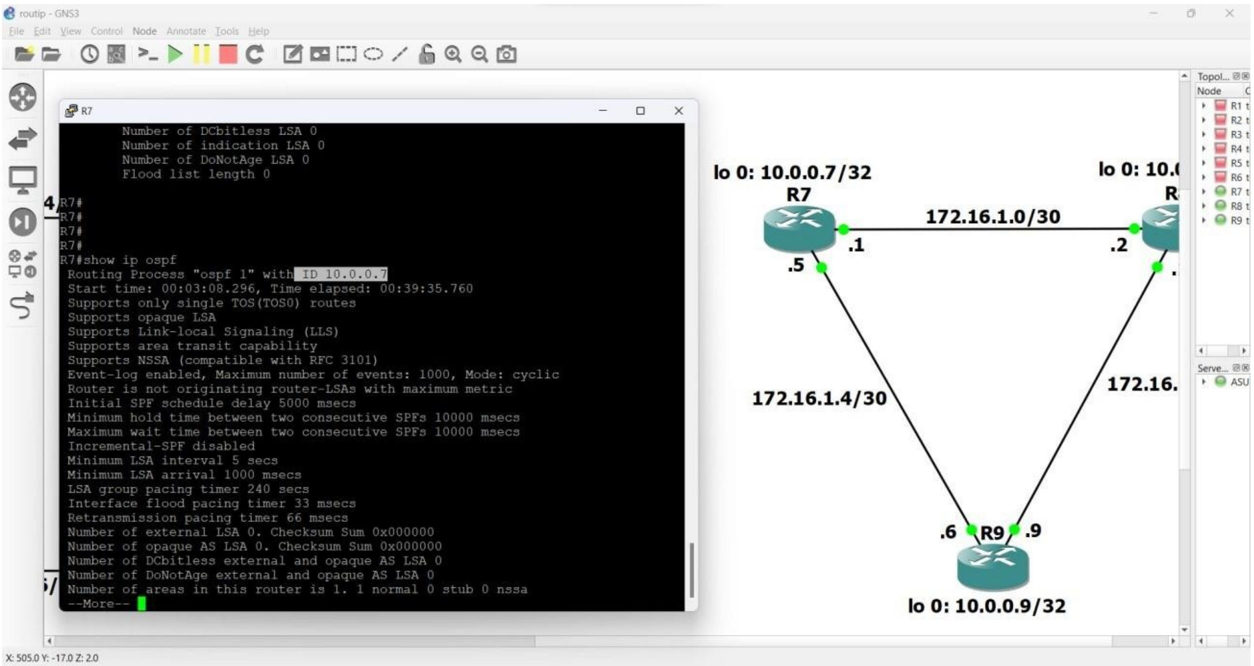


❖ PAQUET HELLO D'OSPF



On peut voir la durée théorique entre deux messages Hello d'OSPF est 10 secondes (Hello interval).

❖ LES IDENTIFIANTS PRIS PAR CHAQUE ROUTEUR



Si un identifiant de routeur n'est pas explicitement configuré, OSPF choisira l'une des adresses IP de ses interfaces comme identifiant de routeur. Il préfère utiliser l'adresse IP de l'interface loopback si elle est configurée, ici il prend l'adresse 10.0.0.7 comme ID de routeur R7.

❖ TEST DE CONNECTIVITE (PING)

- ROUTEUR R7 vers les autres routeurs dans OSPF :

  Connectivité aux routeurs


```
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R7#ping 192.168.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 148/177/212 ms
R7#
*Apr 30 11:09:17.495: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet2/1 from FULL to DOWN, Neighbor Down: Dead ti
mer expired
R7#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/5/8 ms
R7#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/48 ms
R7#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/42/68 ms
R7#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/67/112 ms
R7#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/28/44 ms
R7#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
R7#
```

- ## ROUTEUR R8 :

Ping vers 172.16.8.2
Ping vers 172.16.9.1
Ping vers 172.16.9.2
Ping vers 172.16.7.1
Ping vers 172.16.7.2

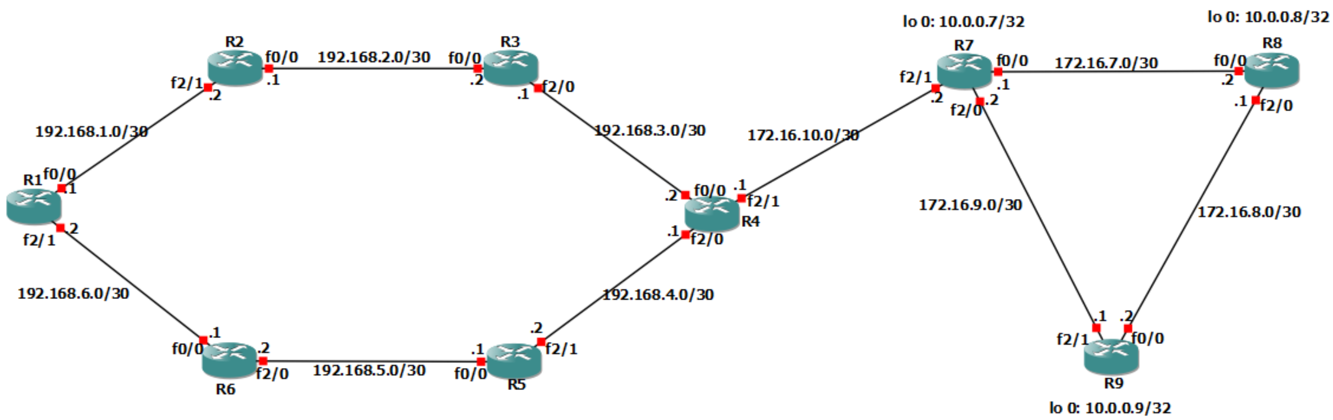
```
R8#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/46/60 ms
R8#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/72/88 ms
R8#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/40/52 ms
R8#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/37/44 ms
R8#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R8#
```

- ROUTEUR R9 :

```
[OK]
R9#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/12 ms
R9#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/48/80 ms
R9#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/71/88 ms
R9#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/55/84 ms
R9#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
R9#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/56/76 ms
R9#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R9#
```

❖ REDISTRIBUTION DES ROUTES ENTRE LES PROTOCOLES RIP et OSPF

- ## Liaison de RIP et OSPF via R4 et R7 :



- ## Vérification des tables de routage dans le réseau RIP

Les tables de routage doivent être bien assuré pour que la configuration soit correcte.

```
      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                     [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/30 is directly connected, FastEthernet2/1
L        192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#
```

❖ ACTIVATION DES NOUVELLES INTERFACES ET ATTRIBUTION D'UNE ADRESSE IP / MASQUE

Ici on va attribuer des adresse IP entre R4 et R7, alors on va configurer et activer OSPF sur ces deux nouvelles interfaces.

- ROUTEUR R4 :

```
*Apr 30 09:12:20.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip add 192.168.3.2 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:08.539: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:16:09.539: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#int f2/0
R4(config-if)#ip add 192.168.4.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:58.587: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:16:59.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R4(config)#int f2/1
R4(config-if)#ip add 172.16.10.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:17:49.963: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:17:50.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
R4(config-router)#
```

- ROUTEUR R7 :

```
R1  R2  R3  R4  R5  R6  R7  R8  R9
R7#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#int lo 0
R7(config-if)#
*Apr 30 09:28:21.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R7(config-if)#ip add 10.0.0.7 255.255.255.255
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#int f2/1
R7(config-if)#ip add 172.16.10.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:30:34.367: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:30:35.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R7(config)#int f0/0
R7(config-if)#ip add 172.16.7.1 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
*Apr 30 09:31:32.599: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:31:33.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7(config-if)#exit
R7(config)#int f2/0
R7(config-if)#ip add 172.16.9.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:32:50.631: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
R7(config)#
*Apr 30 09:32:51.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
solarwinds | Solar-PuTTY free tool          © 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.
```

Après cette configuration et activation des nouvelles interfaces, on va voir si R4 peut pinger l'interface loopback de R8.

❖ PING R4 SUR R8

```
R4#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/108/156 ms
R4#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/68/80 ms
R4#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/73/112 ms
R4#
```

On voit que R4 peut pinger l'interface loopback de R8 car R4 est connecté directement au réseau OSPF via R7 dont R8 est dans le domaine OSPF et aussi parce que les loopback sont déclarées dans OSPF, entrainant les échanges automatiques des routes.

❖ PING DE R8 SUR L'INTERFACE R4 CONNECTE AU ROUTEUR R3

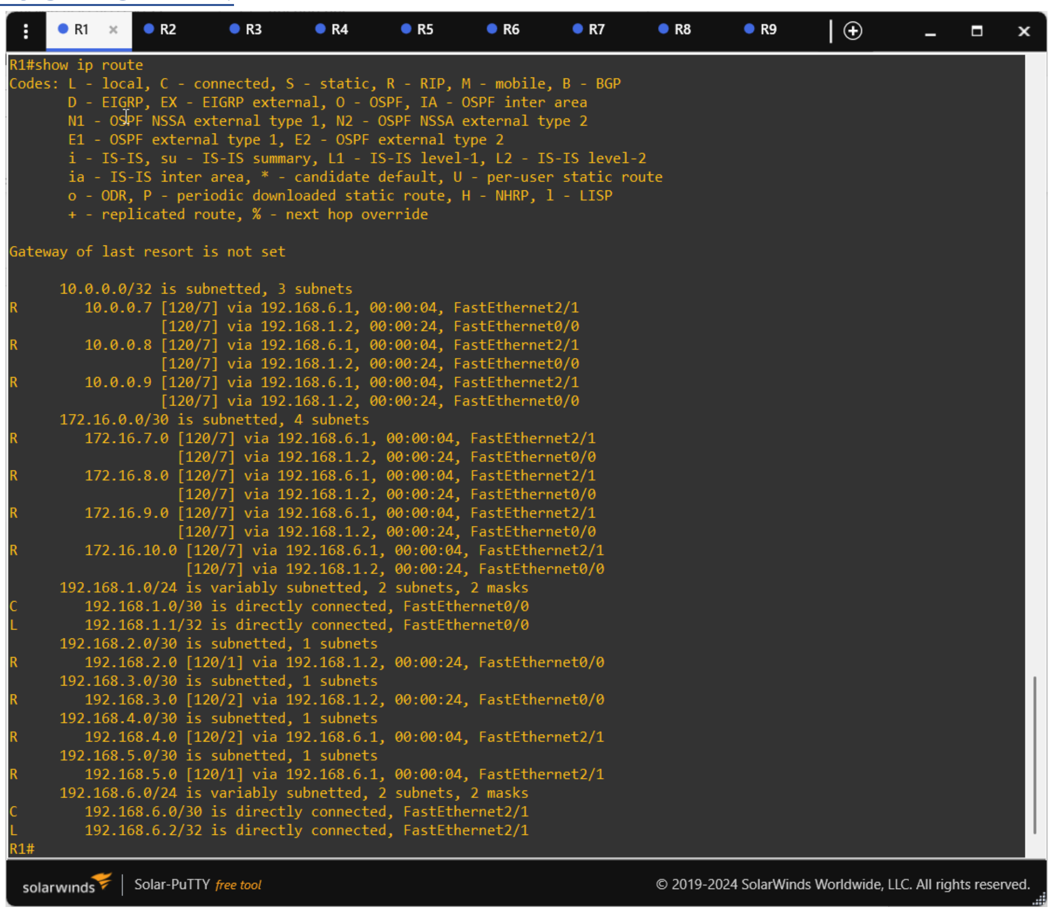
On peut constater que le routeur R8 ne peut pas pinger l'interface de R4 qui est connecté à R3 car les routeurs OSPF n'ont pas configuré le chemin vers les réseaux RIP et aussi que R8 est dans le domaine OSPF tandis que R4 et R3 appartiennent au domaine RIP. De plus, à notre stade là, aucune redistribution n'est encore faite.

❖ PING ENTRE R1 ET R8

Dans ce cas, R1 ne peut pas pinger l'interface de R8 et aussi inversement parce qu'il n'y a pas encore de redistribution entre RIP et OSPF alors qu'aucun chemin valide n'est trouvé entre eux par les routeurs.

❖ TABLE DE ROUTAGE DES ROUTEURS :

- ROUTEUR R1 :




```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R        172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
                     [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/30 is directly connected, FastEthernet2/1
L        192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
```

- ROUTEUR R4 :

```
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
O        10.0.0.7 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
O        10.0.0.8 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
O        10.0.0.9 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O        172.16.7.0/30 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
O        172.16.8.0/30 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
O        172.16.9.0/30 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
C        172.16.10.0/30 is directly connected, FastEthernet2/1
L        172.16.10.1/32 is directly connected, FastEthernet2/1
      192.168.1.0/30 is subnetted, 1 subnets
R        192.168.1.0 [120/2] via 192.168.3.1, 00:00:17, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/1] via 192.168.3.1, 00:00:17, FastEthernet0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, FastEthernet0/0
L        192.168.3.2/32 is directly connected, FastEthernet0/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/30 is directly connected, FastEthernet2/0
L        192.168.4.1/32 is directly connected, FastEthernet2/0
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/1] via 192.168.4.2, 00:00:10, FastEthernet2/0
      192.168.6.0/30 is subnetted, 1 subnets
R        192.168.6.0 [120/2] via 192.168.4.2, 00:00:10, FastEthernet2/0
R4#
```

- ROUTEUR R8 :

❖ REDISTRIBUTION DES ROUTES DE RIP SUR OSPF AU NIVEAU DE R4

Maintenant on va redistribuer les routes de RIP sur OFSP au niveau de R4 :

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
```

Sur le routeur R4, une redistribution est mise en œuvre afin d'intégrer les routes du protocole RIP dans le domaine OSPF, permettant ainsi d'assurer la communication et l'échange d'informations de routage entre ces deux protocoles distincts.

Après la redistribution de route des RIP sur OSPF, une Route vers R1 apparait dans la table de routage de routeur R8 :

✓ TABLE DE ROUTAGE DU ROUTEUR R8 :

```
R8#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
O        10.0.0.7 [110/2] via 172.16.7.1, 02:03:00, FastEthernet0/0
C        10.0.0.8 is directly connected, Loopback0
O        10.0.0.9 [110/2] via 172.16.8.2, 01:57:01, FastEthernet2/0
      172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C        172.16.7.0/30 is directly connected, FastEthernet0/0
L        172.16.7.2/32 is directly connected, FastEthernet0/0
C        172.16.8.0/30 is directly connected, FastEthernet2/0
L        172.16.8.1/32 is directly connected, FastEthernet2/0
O        172.16.9.0/30 [110/2] via 172.16.8.2, 01:57:01, FastEthernet2/0
                       [110/2] via 172.16.7.1, 02:03:00, FastEthernet0/0
O        172.16.10.0/30 [110/2] via 172.16.7.1, 00:42:25, FastEthernet0/0
      192.168.1.0/30 is subnetted, 1 subnets
O E2     192.168.1.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
O E2     192.168.2.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
O E2     192.168.3.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
O E2     192.168.4.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
      192.168.5.0/30 is subnetted, 1 subnets
O E2     192.168.5.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
      192.168.6.0/30 is subnetted, 1 subnets
O E2     192.168.6.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
R8#
```

Alors on va effectuer un ping de R8 vers une interface de R1 ;

❖ FONCTIONNEMENT DU PING DE R8 Vers UNE INTERFACE DE R1 :

```
R8#ping 192.168.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 148/156/164 ms
R8#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 128/149/168 ms
R8#
```

D'après ce qu'on voit, on peut déterminer que le ping de R8 vers une interface de R1 fonctionne très bien.

• Paquets ICMP :

Le paquet ICMP arrive à R1 mais R1 ne peut pas envoyer des réponses car le chemin vers les réseaux OSPF n'existe pas dans la table de routage de R1 (retour non garanti).

❖ PING DE R1 VERS R8 :

Le routeur R1 peut maintenant pinger le routeur R8 parce que la redistribution est déjà faite entre RIP et OSPF.

❖ REDISTRIBUTION DES TABLES DE ROUTAGE OSPF SUR RIP AU NIVEAU DE R4

Ici, on va redistribuer les tables de routage OSPF sur RIP au niveau de R4.

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
```

Et puis regardons l'une des tables de routage des routeurs RIP, des modifications doivent y parvenir :

```
Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        10.0.0.8 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        10.0.0.9 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.8.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.9.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R        172.16.10.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet2/1
L        192.168.1.2/32 is directly connected, FastEthernet2/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, FastEthernet0/0
L        192.168.2.1/32 is directly connected, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/1] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.2.2, 00:00:05, FastEthernet0/0
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/2] via 192.168.1.1, 00:00:09, FastEthernet2/1
      192.168.6.0/30 is subnetted, 1 subnets
R        192.168.6.0 [120/1] via 192.168.1.1, 00:00:09, FastEthernet2/1
R2#
```

On a pris la table de routage du routeur R2 comme exemple.

❖ APPARITION DES ROUTES

Les routes vers OSPF n'apparaissent plus sur les tables de routage des routeurs RIP car la valeur de métrique par défaut vers les réseaux OSPF est plus élevé que les métriques dans les réseaux RIP or le protocole ne considère pas cette Route comme plus court chemin.

✓ PING de R1 à R8 :

```
R1#
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 208/244/336 ms
R1#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 180/200/232 ms
R1#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 180/216/248 ms
R1#
```

❖ CORRECTION DU PROBLEME

Il faut définir une métrique pour que RIP sache que quelle métrique il doit appliquer parce que c'est la cause du problème (routes ignorées et incomplètes).
Cela permet à RIP d'accepter les routes OSPF et de les propager correctement.

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
```

• VERIFICATION :

Affichons une des tables de routage des réseaux RIP pour bien vérifier si le problème est résolu :

Test PING :

```
R8#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 136/152/168 ms
R8#ping 192.168.6.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 152/167/188 ms
R8#
```

```
R1#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 144/152/160 ms
R1#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 188/222/268 ms
R1#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 184/196/204 ms
R1#
```

Table de routage d’un Routeur :

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 3 subnets
R        10.0.0.7 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R        10.0.0.8 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R        10.0.0.9 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                  [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
      172.16.0.0/30 is subnetted, 4 subnets
R        172.16.7.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R        172.16.8.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R        172.16.9.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                    [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R        172.16.10.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
                     [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
      192.168.2.0/30 is subnetted, 1 subnets
R        192.168.2.0 [120/1] via 192.168.1.2, 00:00:02, FastEthernet0/0
      192.168.3.0/30 is subnetted, 1 subnets
R        192.168.3.0 [120/2] via 192.168.1.2, 00:00:02, FastEthernet0/0
      192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/2] via 192.168.6.1, 00:00:13, FastEthernet2/1
      192.168.5.0/30 is subnetted, 1 subnets
R        192.168.5.0 [120/1] via 192.168.6.1, 00:00:13, FastEthernet2/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/30 is directly connected, FastEthernet2/1
L        192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
```

Après la résolution du problème, les communications entre les routeurs doivent être fonctionnelles :

- Toutes les Routes vers les réseaux OSPF apparaissent sur les tables de routage des routeurs RIP.
- Le test de connexion en effectuant des pings est réussi.